

Algebra PhD Exam, September 2009

Do seven of the following eleven problems. Do not turn in more than seven problems. You must show your work. Answers with no work and/or no explanations will receive no credit. State clearly any theorem you use in your proofs.

In the problems, $\mathbb{Z}$, resp. $\mathbb{Q}$, $\mathbb{C}$, is the set of all integers, resp. of all rational numbers, of all complex numbers.

1. State and prove the Jordan–Hölder theorem for finite groups.
2. This problem has two parts.
 - (a) Show that the additive group of a field is never a free abelian group.
 - (b) Show that the multiplicative group of positive rational numbers is a free abelian group, and find a basis.
3. A category C is small if its class of objects is a set, and skeletal if for all X and Y in C , $X \simeq Y$ implies $X = Y$. Show that any small category is equivalent to a skeletal one (choose an object from each isomorphism class).
4. Let R be a commutative ring.
 - (a) Define what it means for an R -module M to be flat.
 - (b) Show that the polynomial ring $R[x]$, considered as an R -module in the natural way, is flat.
5. Show that if R is a commutative ring with identity and every prime ideal of R is finitely generated, then R is Noetherian.
6. Show that for any extension $A \rightarrow B$ of commutative rings, any A -module M and any B -module N , there is an isomorphism
$$\mathrm{Hom}_B(M \otimes_A B, N) \simeq \mathrm{Hom}_A(M, N_A)$$
functorial in M and N (here N_A denotes N considered as an A -module by means of the ring homomorphism $A \rightarrow B$).
7. This problem has two parts.
 - (a) Show that if L/F , K/L are finite separable extensions, then so is K/F .
 - (b) Do the same for purely inseparable extensions.
8. This problem has three parts.
 - (a) Show that if K/F is a normal extension and $F \subseteq L \subseteq K$ is an intermediate field, then K/L is normal.
 - (b) Give an example where L/F is not normal.
 - (c) Give an example of normal extensions L/F , K/L where K/F is not normal.
9. **Warning: Suspected error.** The source says only that p is prime, but the assertion that K contains a quadratic extension of $\mathbb{Q}$ fails when $p = 2$. A possible omitted restriction cannot be determined uniquely, so the source's mathematical scope has been retained.

Let p be a prime and K be the splitting field over $\mathbb{Q}$ of $X^p - 1$.

- (a) Show that K contains a unique quadratic extension of $\mathbb{Q}$.
- (b) When is this quadratic extension real? (You may assume as known the structure of the Galois group of $K/\mathbb{Q}$. hint: if ζ is a primitive p th root of unity, complex conjugation sends $\zeta \mapsto \zeta^{-1}$).

10. Warning: Suspected error. Even after the grammatical correction, the printed assertion appears to require an additional hypothesis: a finite-dimensional semisimple k -algebra need not be a direct sum of fields. No missing hypothesis is uniquely determined, so none has been added.

Let k be a field and R a finite-dimensional k -algebra. Show that if R is semisimple as a ring, it is a direct sum of extension fields of k .

11. Classify up to isomorphism all semisimple rings with 720 elements.